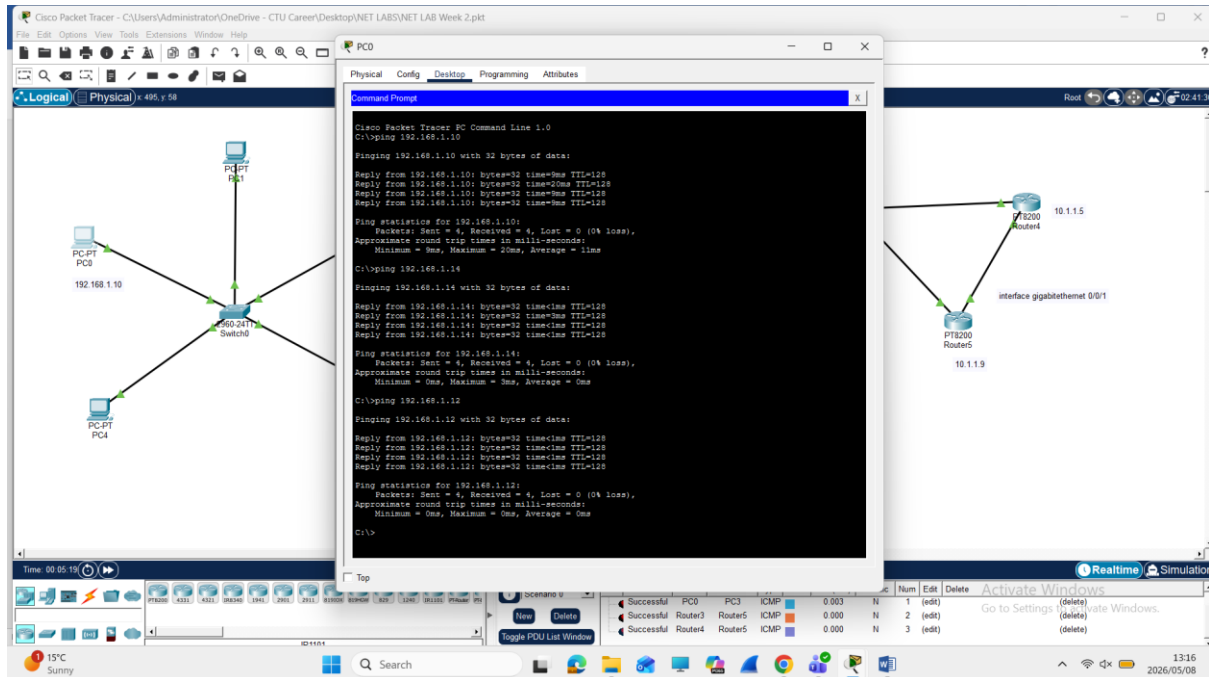
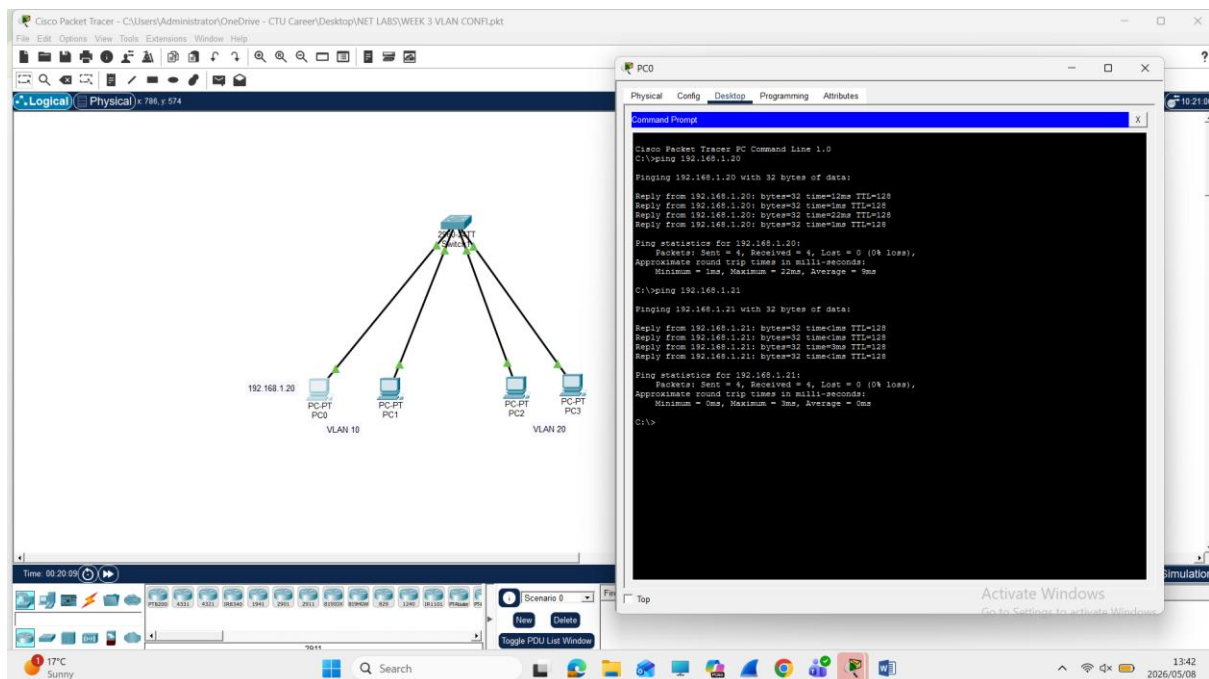


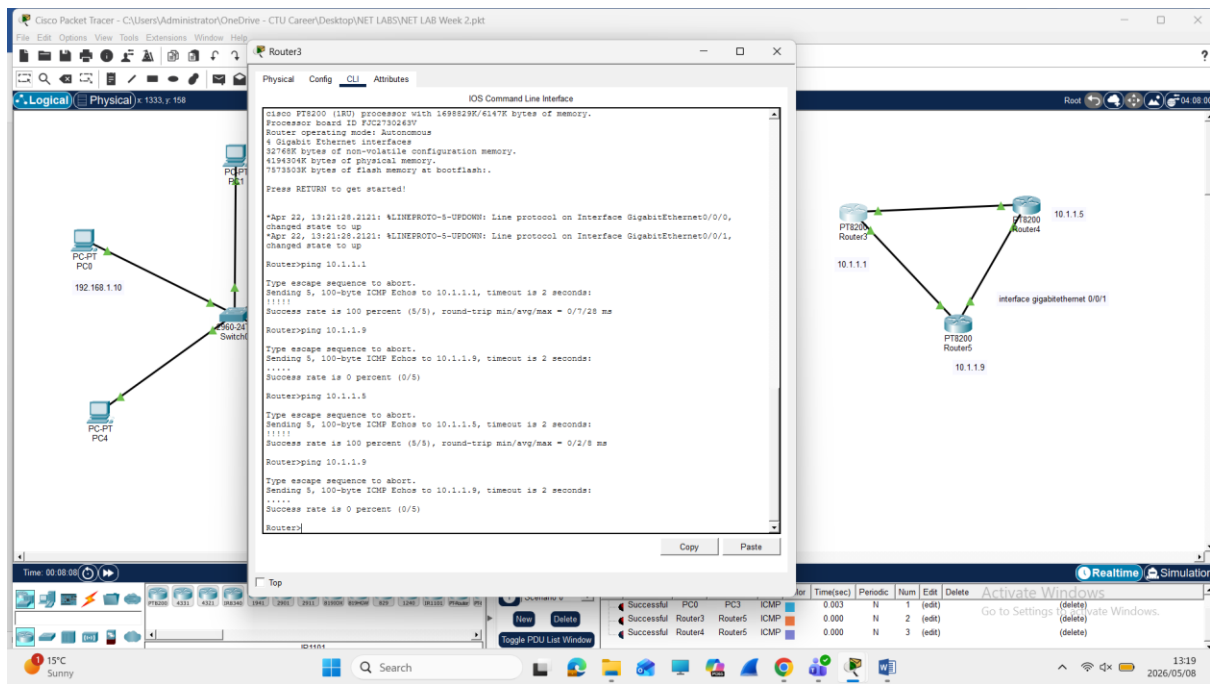
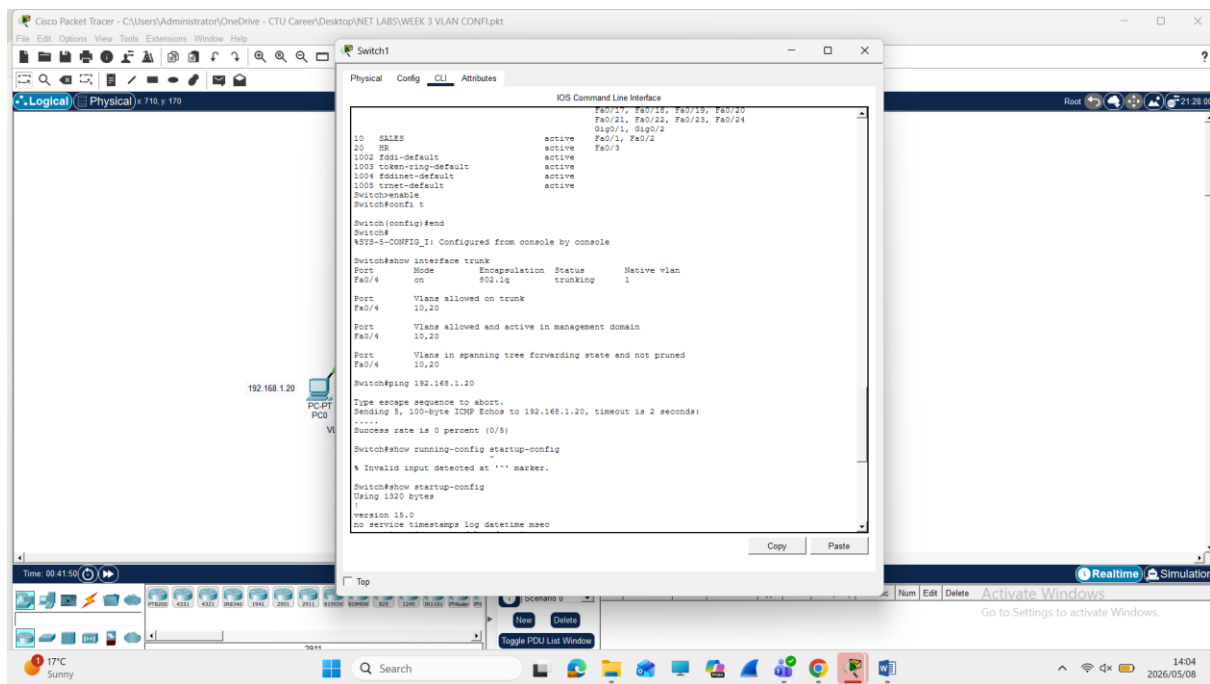
Week 2



Redundancy and fault tolerance assessment are done to ensure that a network can continue working or if a connection fails. Ping test can help check connectivity for if there may be a loss of files that may need to be deliver to another network, it shows that a network is working correctly and that everything working effectively. Protocols are set in place and need to be configured in correctly so that they can provide automatic failovers if a router in a topology. Assessments can show weather a connection is set up correctly or if something needs to be configured or connected correctly

Week3





Week 4

The screenshot shows the Cisco Packet Tracer interface. The main window displays the configuration for Router0. The configuration includes the following commands:

```
Router>ipconfig/all
Translating "ipconfig/all"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router>show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.1.1 YES manual up up
GigabitEthernet0/0/1 unassigned YES unset down down
GigabitEthernet0/0/2 unassigned YES unset down down
GigabitEthernet0/0/3 unassigned YES unset down down
Vlan1 unassigned YES unset up down

Router>show ipv6 interface brief
Router>show ipv6 interface brief
GigabitEthernet0/0/0 [up/up]
FE80::290:CFF:FE8B:1B01
2001:DB8:111:10
GigabitEthernet0/0/1 [up/down]
unassigned
GigabitEthernet0/0/2 [up/down]
unassigned
GigabitEthernet0/0/3 [up/down]
unassigned
Vlan1 [up/down]
unassigned
Router>show ipv6 interface brief
% Invalid input detected at '^' marker.

Router>show ip route
Codes: C - connected, S - static, I - IGMP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L      192.168.1.1/32 is directly connected, GigabitEthernet0/0/0

Router#
```

The network diagram at the bottom shows Router0 connected to PC0 and PC1. The status of the connections is shown in the table below:

Num	Edit	Delete	Activate Windows
0	(edit)	(delete)	(delete)
1	(edit)	(delete)	(delete)

The screenshot shows the Cisco Packet Tracer interface. The main window displays the configuration for Router0. The configuration includes the following commands:

```
GigabitEthernet0/0/3 [up/down]
Vlan1 [up/down]
unassigned
Router>show ipv6 interface brief
% Invalid input detected at '^' marker.

Router>show ip route
Codes: C - connected, S - static, I - IGMP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L      192.168.1.1/32 is directly connected, GigabitEthernet0/0/0

Router>ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echoes to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/8/15 ms

Router>ping 2001:db8:111:10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echoes to 2001:db8:111:10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

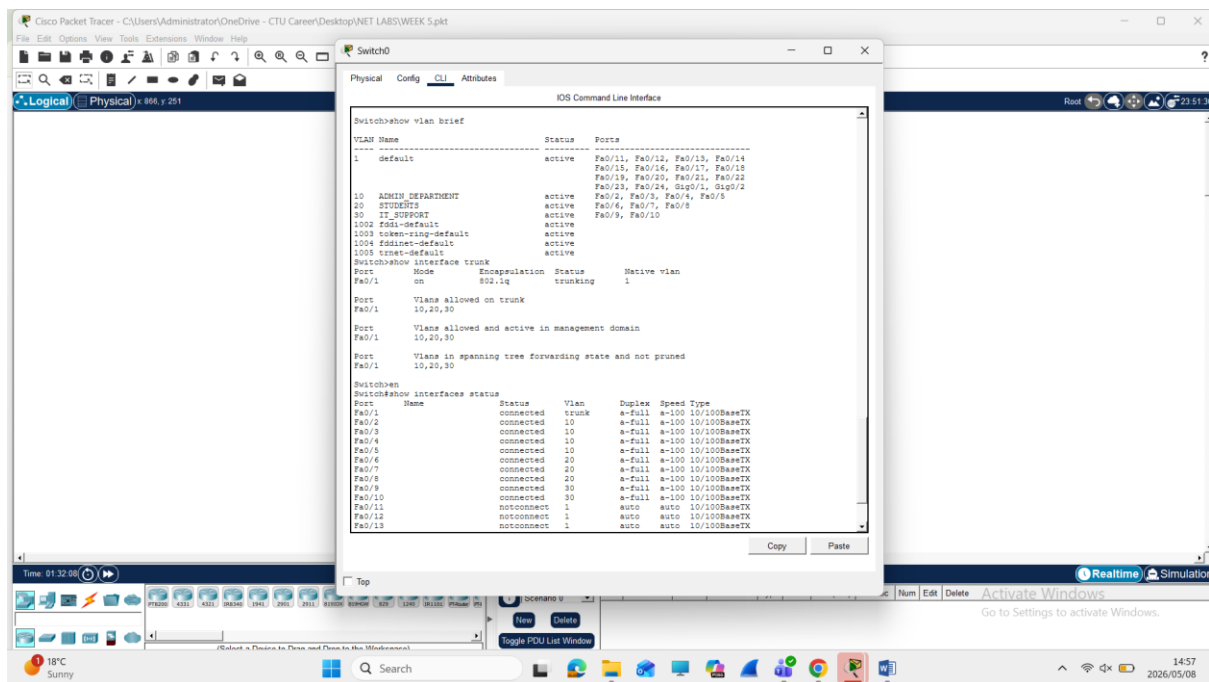
Router>ping 2001:db8:111:10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echoes to 2001:db8:111:10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/5/7 ms

Router#
```

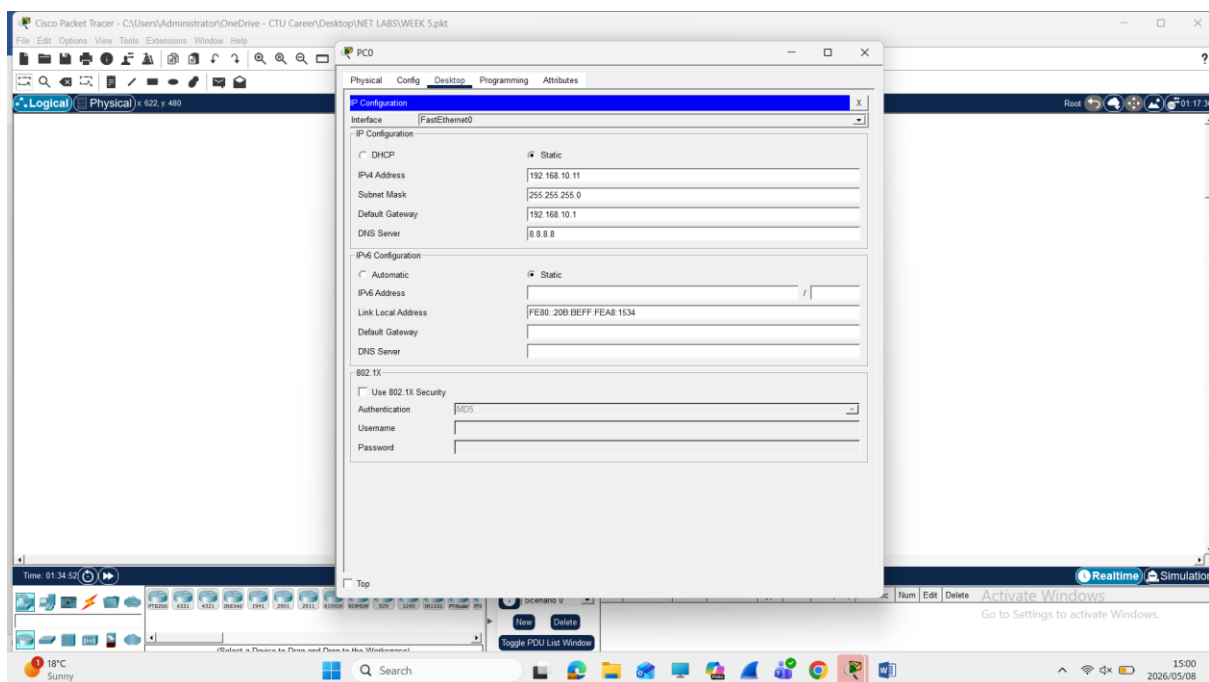
The network diagram at the bottom shows Router0 connected to PC0 and PC1. The status of the connections is shown in the table below:

Num	Edit	Delete	Activate Windows
0	(edit)	(delete)	(delete)
1	(edit)	(delete)	(delete)

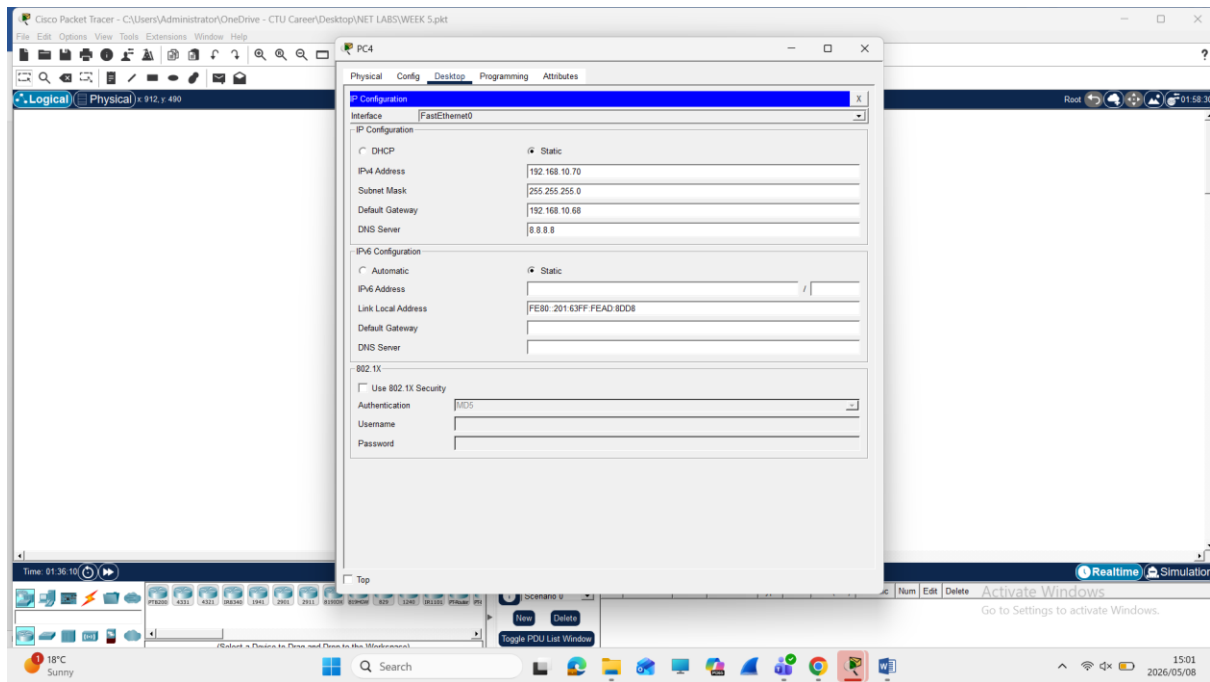
Week 5



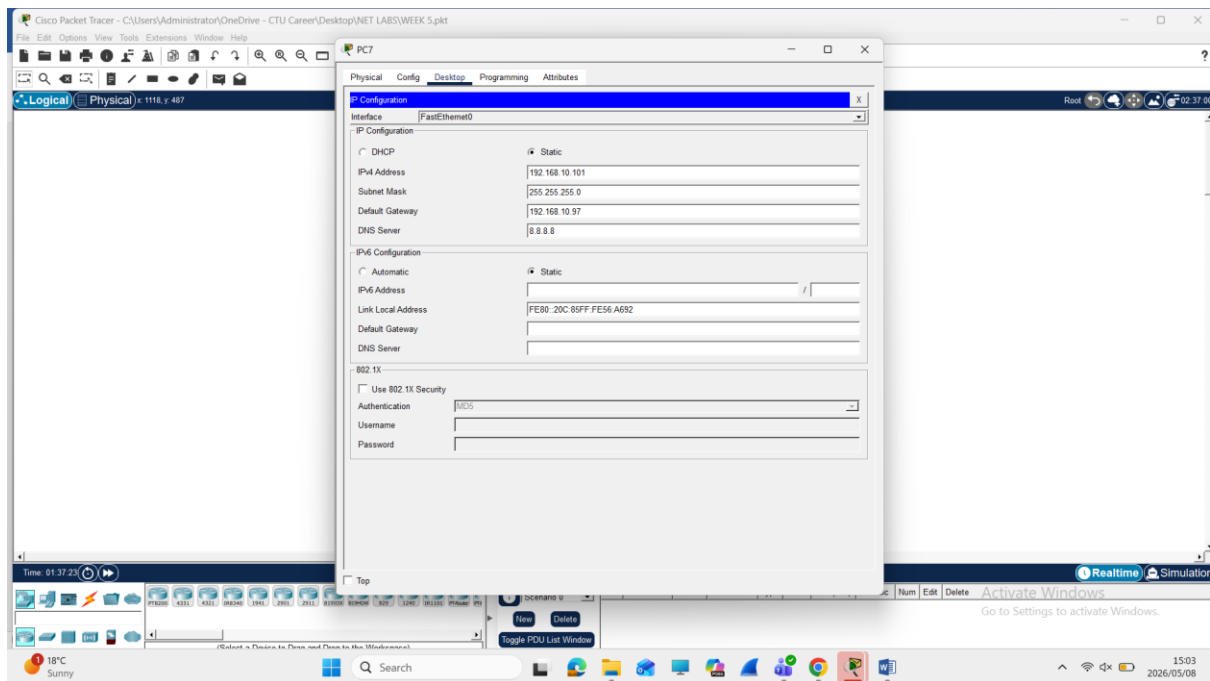
Admin

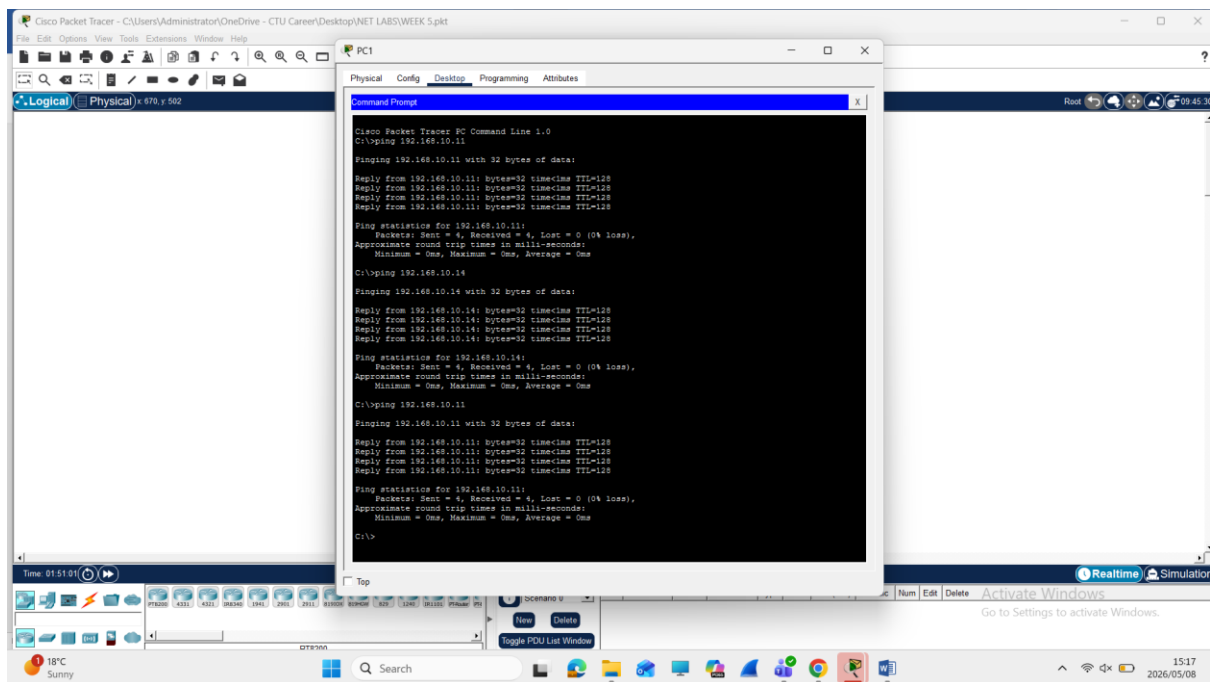
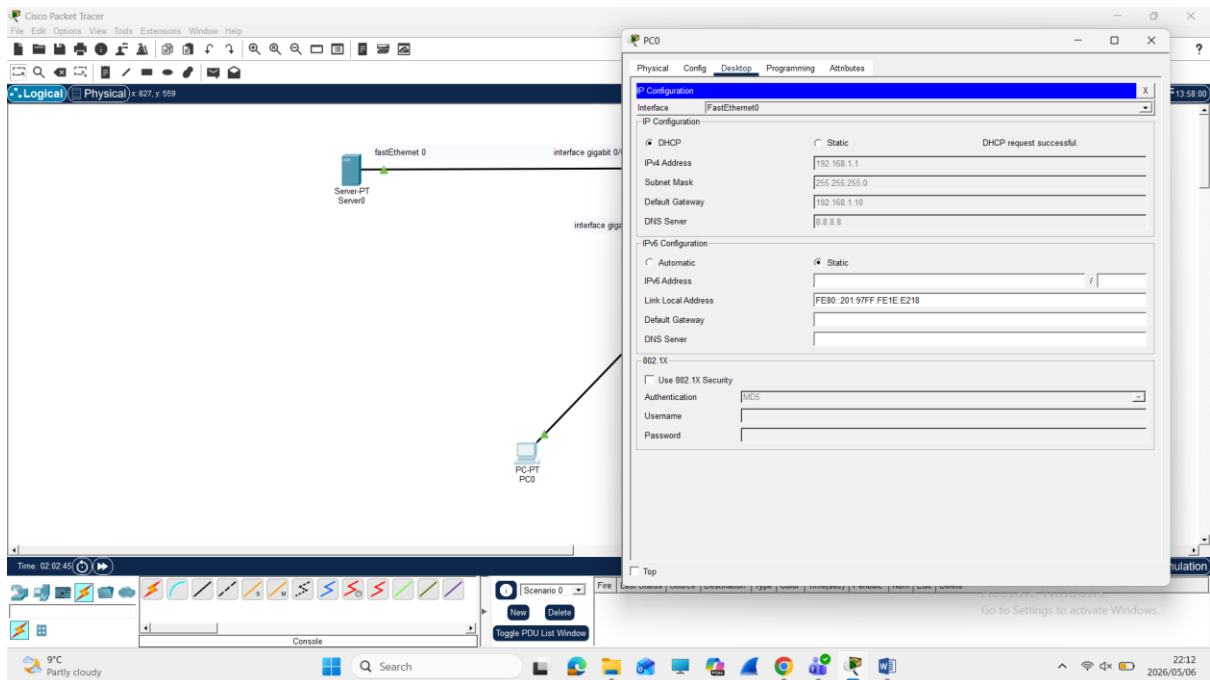


Students



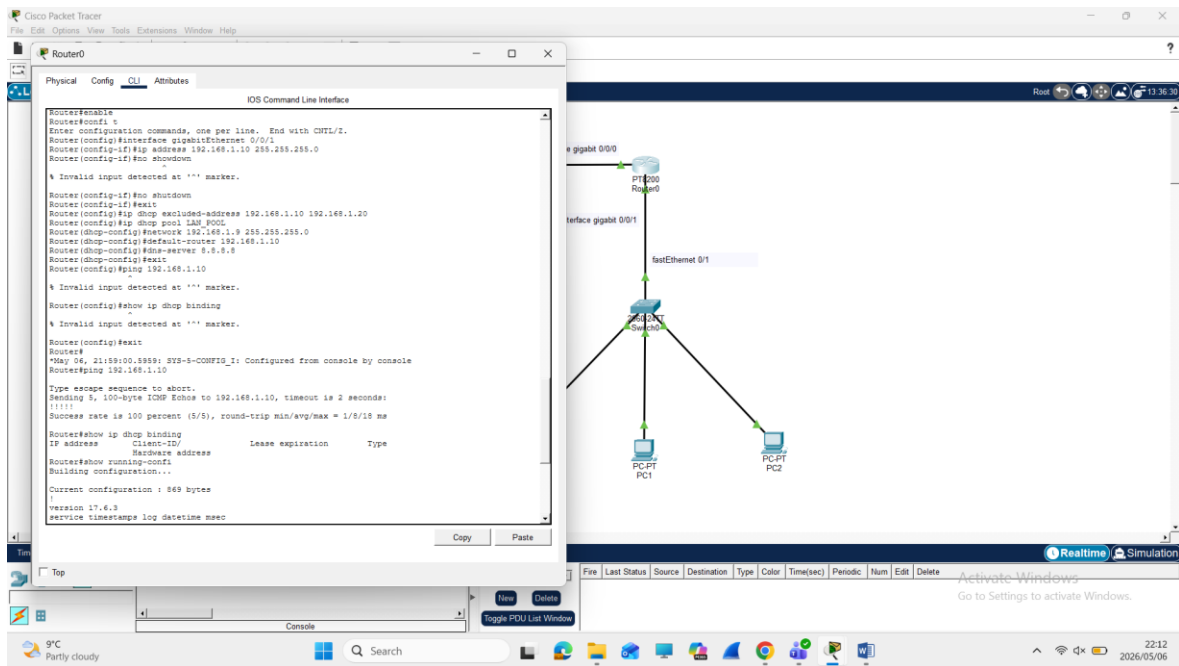
IT Support



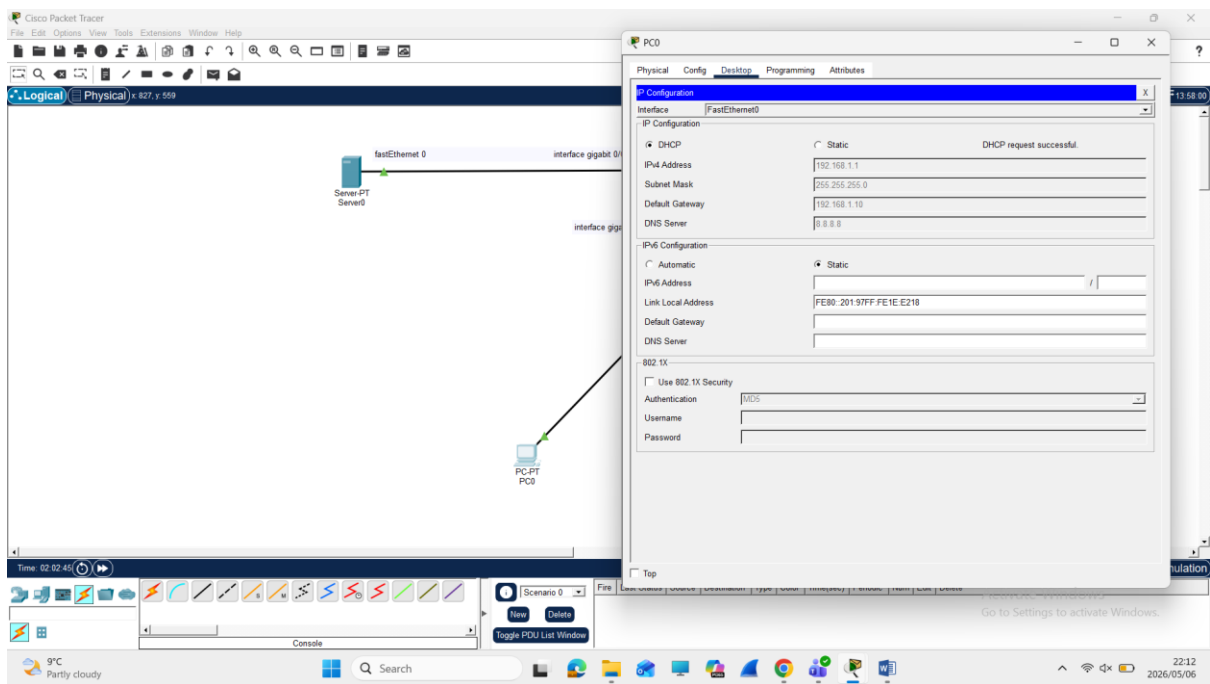


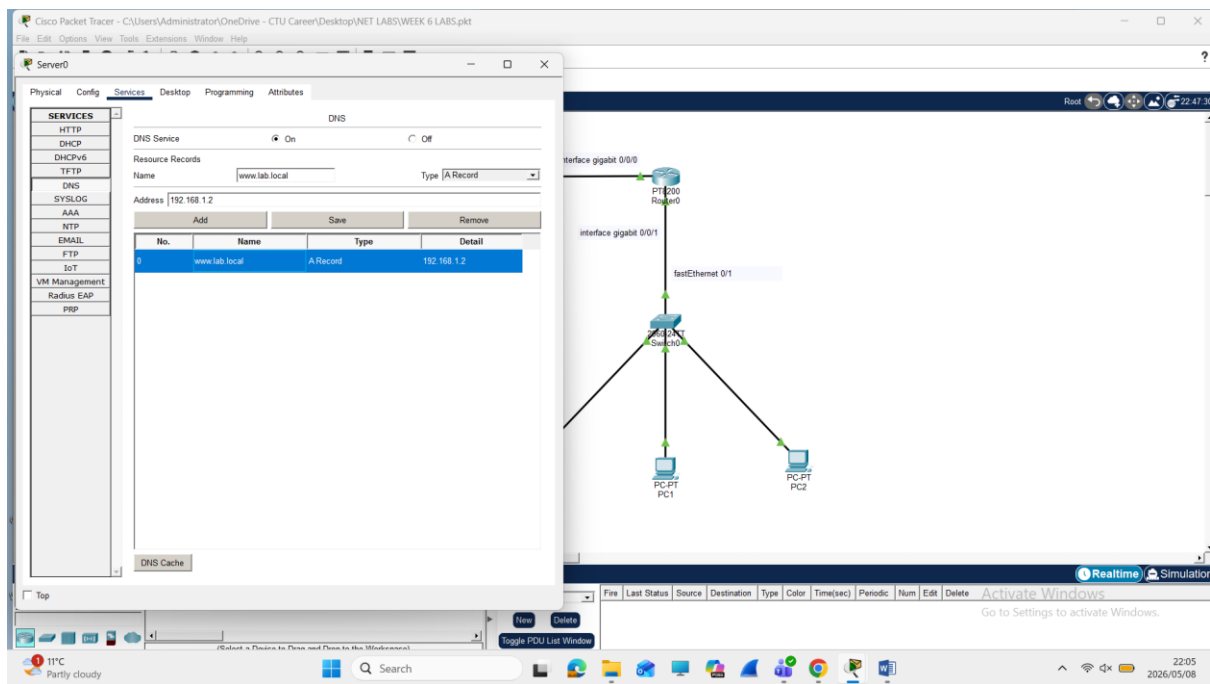
Week 6

1

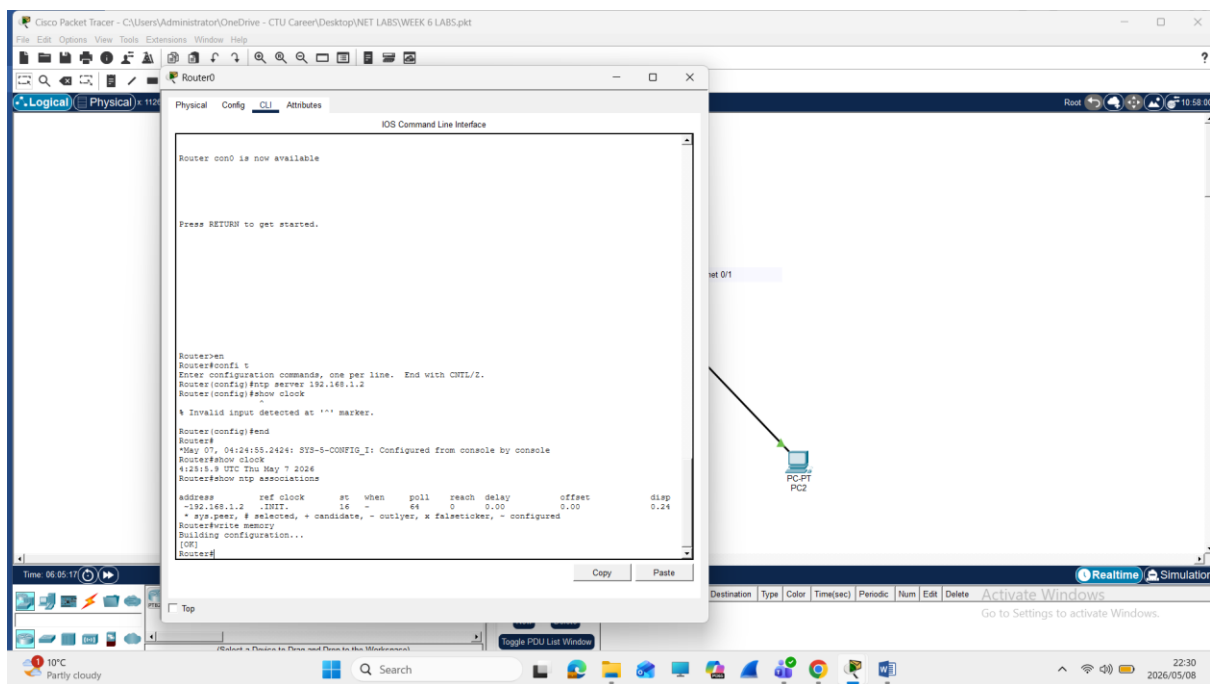


2





3



5

Cisco Packet Tracer - C:\Users\Administrator\OneDrive - CTU Career\Desktop\NET LABS\WEEK 6 LABS.pkt

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```

Router>en
Router#confi t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname r1
r1(config)#ip domain-name cisco.com
% Invalid input detected at '^' marker.

r1(config)#crypto key generate rsa
% Please define a domain-name first.
r1(config)#cisco.com
% Invalid input detected at '^' marker.

r1(config)#ip domain-name cisco.com
r1(config)#crypto key generate rsa
The name for the keys will be: r1.cisco.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

r1(config)#ip ssh version 2
*May 7 12:50:58.818: %SSH-5-ENABLED: SSH 1.99 has been enabled
r1(config)#username admin secret 1234
% Invalid input detected at '^' marker.

r1(config)#username admin secret 1234
r1(config)#line vty 0 4
r1(config-line)#login local
r1(config-line)#transport input ssh
r1(config-line)#exit
r1(config)#end
r1#
*May 07, 12:56:25.5656: SYS-5-CONFIG_I: Configured from console by console
r1#write memory
Building configuration...
[OK]
r1#

```

Time: 07

Copy Paste

Toggle PDU List Window

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

WEEK 7

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x 608, y 217

Time: 00:51:54

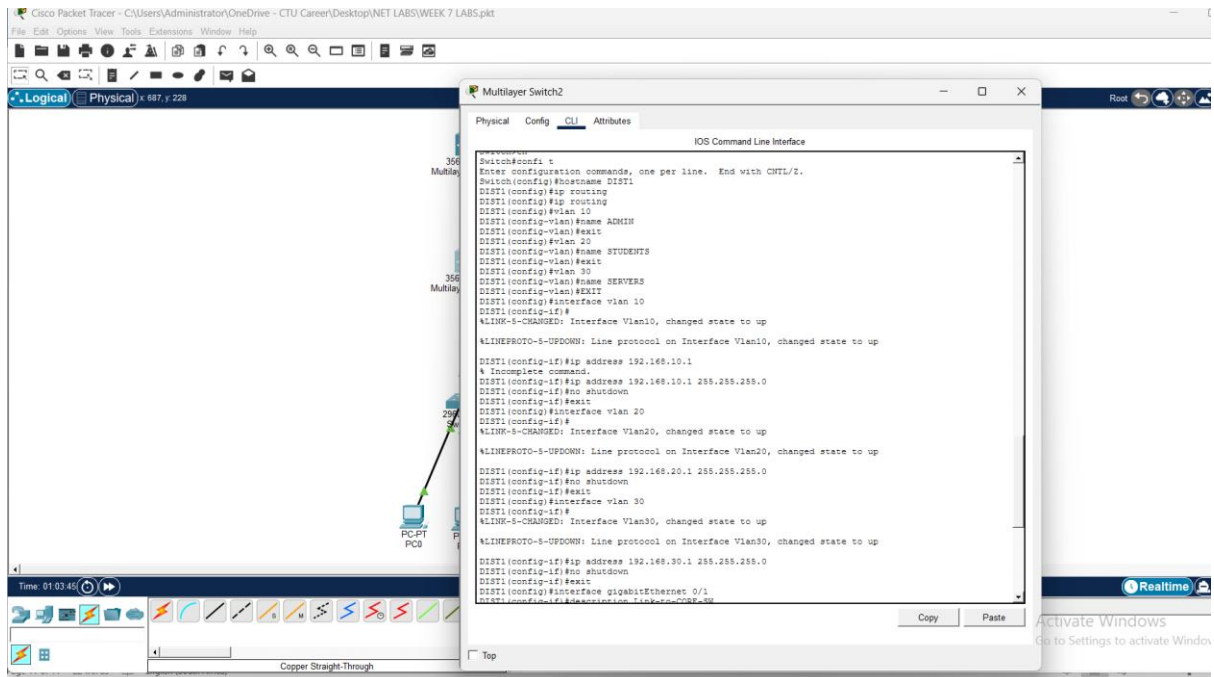
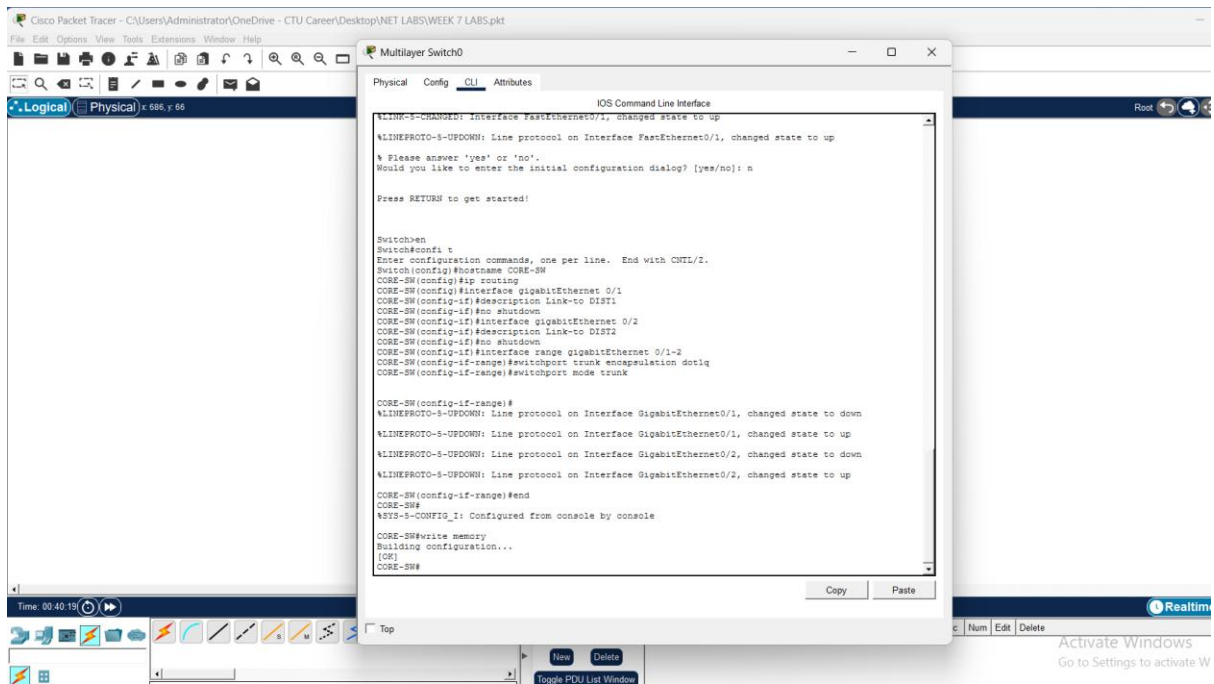
Scenario 0

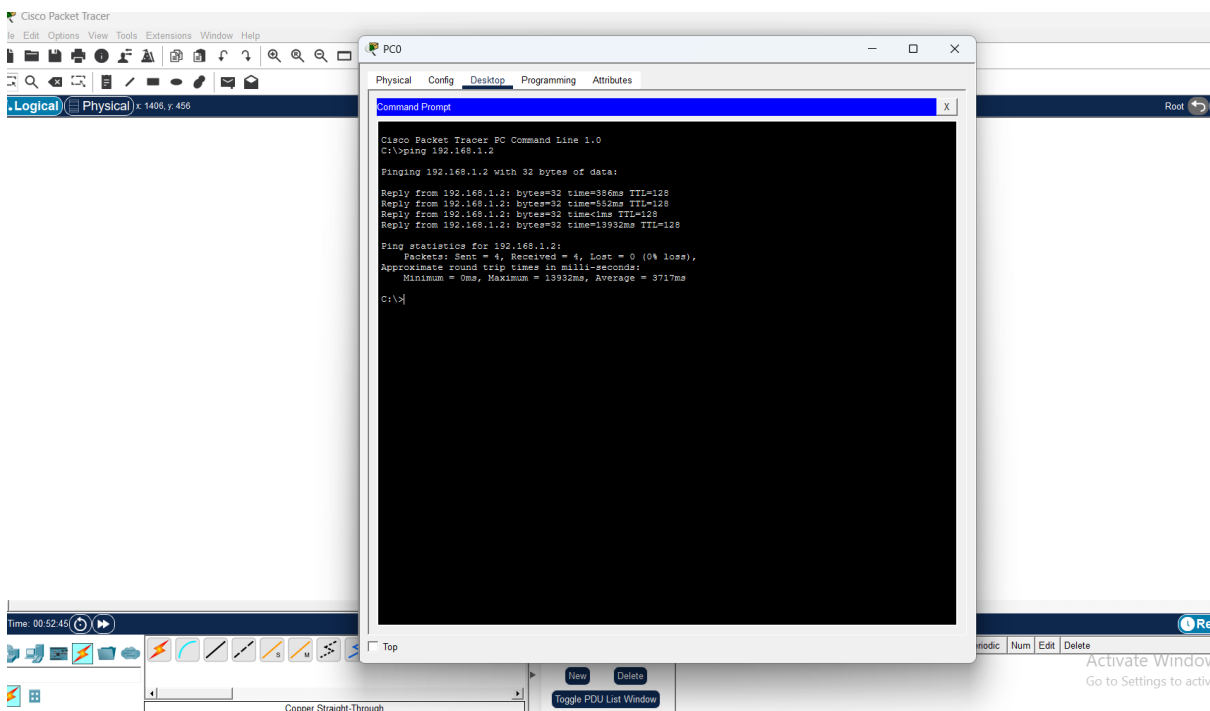
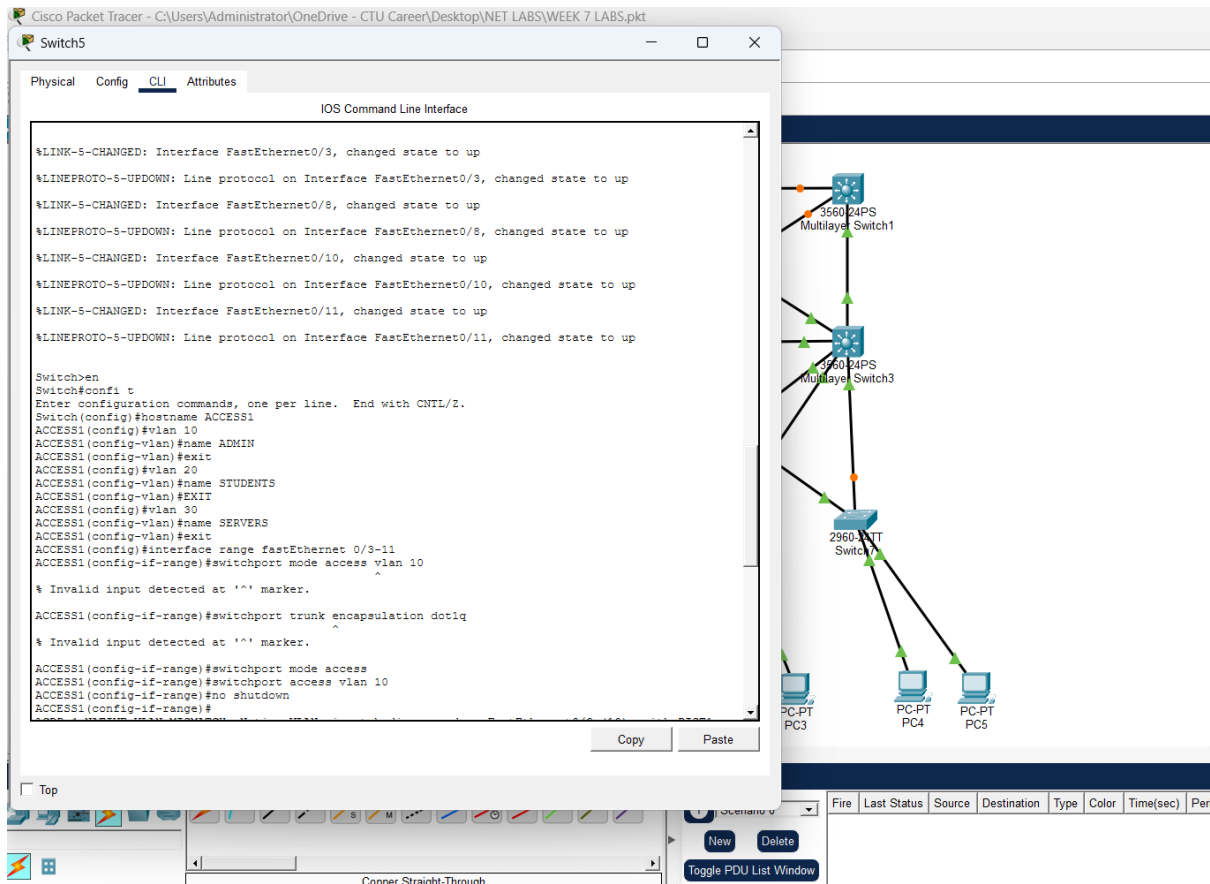
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

Activate Go to Settings

Toggle PDU List Window

Copper Straight-Through





HSRP/VRRP are a Cisco protocol that keep networks running when a router fails. In a topology if there are two routers one works as the main router and the second one is used as the backup router. They can share the same IP address and the same default gateway for encase the main stops working the backup ac automatically take over so that network can stay connected

WEEK 8

Logical Physical x: 1292, y: 672

```
graph LR; Router0[PT8200 Router0] --- Switch0[2960 24TT Switch0]; Switch0 --- PC0[PC-PT PC0];
```

Time: 26:55:33

Scenario 0

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

Activate Wi Go to Settings t

(Select a Device to Drag and Drop to the Workspace)

Cisco Packet Tracer - C:\Users\Administrator\OneDrive - CTU Career\Desktop\NET LABS\WEEK 8 LABS.pkt

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>EN
Router>confi t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface gigabitEthernet 0/0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#enable secret cisco123
R1(config)#line console 0
R1(config-line)#password cisco123
R1(config-line)#login
R1(config-line)#line vty 0 4
R1(config-line)#password cisco123
R1(config-line)#login
R1(config-line)#end
R1#
*May 15, 21:38:49.3838: SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#vr erase
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
R1#reload
Proceed with reload? [confirm]
Initializing Hardware ...
U
Checking for PCIe device presence...done
Reading confreg 8000
System integrity status: 0x610

Key Sectors: (Primary, GOOD), (Backup, GOOD), (Revocation, GOOD)
Size of Primary = 2452 Backup = 2452 Revocation = 1008
RSA Self Test Passed

Expected hash:
D0AF35A193617ABACC417349AE204131
12E6FA4E89A97EA20A9EE264B5D09A
12E6FA4E89A97EA20A9EE264B5D09A
```

Copy Paste

Time: 00:00:00

Top

Copper Straight-Through

New Delete

Toggle PDU List Window

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

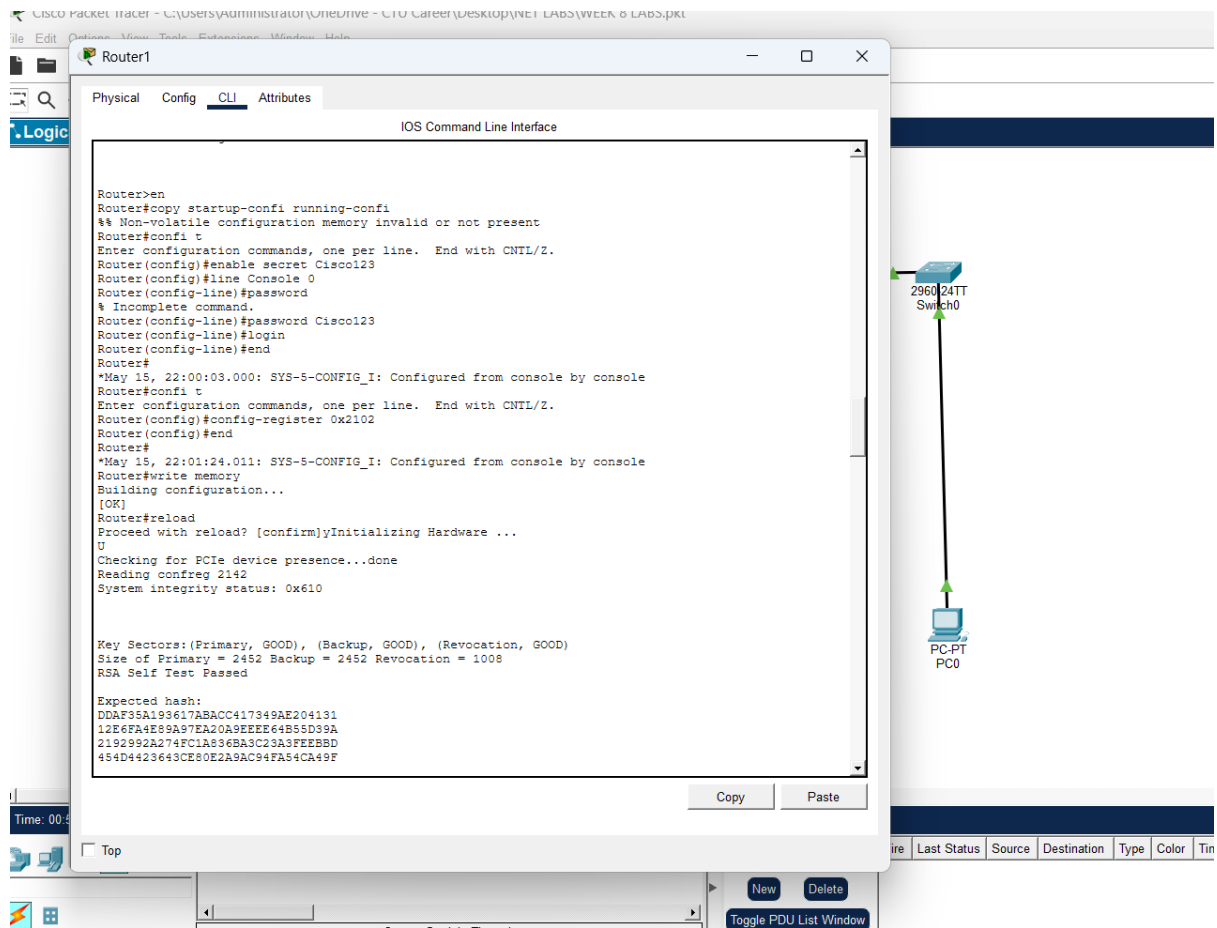
Factory reset

The screenshot shows the Cisco Packet Tracer interface. The main window displays Router1 in the CLI (Command Line Interface) mode. The CLI window shows the output of the 'reload' and 'reset' commands, indicating a successful factory reset. The output includes system boot information, memory details, and a successful ROM image verification.

```
System Bootstrap, Version 17.6(6r), RELEASE SOFTWARE  
Copyright (c) 1994-2022 by cisco Systems, Inc.  
  
Current image running : Boot ROM0  
  
Last reset cause : LocalSoft  
PT8200 platform with 4194304 Kbytes of main memory  
  
Initializing memory for ECC  
..  
PT8200 processor with 524288 Kbytes of main memory  
Main memory is configured to 64 bit mode with ECC enabled  
  
rommon 1 > reload  
monitor: command "reload" not found  
rommon 2 > reset  
Initializing Hardware ...  
U  
Checking for PCIE device presence...done  
Reading confreg 8000  
System integrity status: 0x610  
  
Key Sectors:(Primary, GOOD), (Backup, GOOD), (Revocation, GOOD)  
Size of Primary = 2452 Backup = 2452 Revocation = 1008  
RSA Self Test Passed  
  
Expected hash:  
DDAF35A193617ABACC417349AE204131  
12E6FA4E89A97EA20A9EEEE64B55D39A  
2192992A274FC1A836BA3C23A3FEEBBD  
454D4423643CE80E2A9AC94FA54CA49F  
  
Obtained hash:  
DDAF35A193617ABACC417349AE204131  
12E6FA4E89A97EA20A9EEEE64B55D39A  
2192992A274FC1A836BA3C23A3FEEBBD  
454D4423643CE80E2A9AC94FA54CA49F  
Sha512 Self Test Passed  
Rom image verified correctly
```

The background shows a network diagram with a 2960 24TT Switch0 connected to a PC-PT PC0. The bottom status bar indicates 'Copper Straight-Through' and 'Toggle PDU List Window'.

Password recovery



Tftp

Router0

Physical Config CLI Attributes

Vlan1unassignedYES NVRAMadministratively down down

Router#config t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface g0/0

Router(config-if)#ip address 192.168.1.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#ex

Router(config)#ex

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.1.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	NVRAM	administratively down	down
GigabitEthernet0/2	unassigned	YES	NVRAM	administratively down	down
Vlan1	unassigned	YES	NVRAM	administratively down	down

Router#ping 192.168.1.100

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.100, timeout is 2 seconds:

..!!!!

Success rate is 80 percent (4/5), round-trip min/avg/max = 1/6/12 ms

Router#copy running-config tftp

Address or name of remote host []? 192.168.1.100

Destination filename [Router-config]?

Writing running-config...!!

[OK - 709 bytes]

709 bytes copied in 0.013 secs (54538 bytes/sec)

Router#copy startup-config tftp

Address or name of remote host []? 192.168.1.100

Destination filename [Router-config]?

Writing startup-config...!!

[OK - 776 bytes]

776 bytes copied in 0 secs

Router#

ios backup

Router0

Physical Config CLI Attributes

```
rommon 5 > TFTP_FILE=c1900-universalk9-mz.SPA155-3.M4a.bin
rommon 6 > tftpdnld

      IP_ADDRESS: 192.168.1.1
      IP_SUBNET_MASK: 255.255.255.0
      DEFAULT_GATEWAY: 192.168.1.254
      TFTP_SERVER: 192.168.1.100
      TFTP_FILE: c1900-universalk9-mz.SPA155-3.M4a.bin
Invoke this command for disaster recovery only.
WARNING: all existing data in all partitions on flash will be lost!

Do you wish to continue? y/n:  [n]:  y

.....[TIMED OUT]
TFTP: Operation terminated.

rommon 7 > FE_SPEED_MODE=4
rommon 8 > tftpdnld

      IP_ADDRESS: 192.168.1.1
      IP_SUBNET_MASK: 255.255.255.0
      DEFAULT_GATEWAY: 192.168.1.254
      TFTP_SERVER: 192.168.1.100
      TFTP_FILE: c1900-universalk9-mz.SPA155-3.M4a.bin
Invoke this command for disaster recovery only.
WARNING: all existing data in all partitions on flash will be lost!

Do you wish to continue? y/n:  [n]:  y

.....[TIMED OUT]
TFTP: Operation terminated.

rommon 9 > System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO2911/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 72/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized
```

Copy Paste

Top